Sales Data Analysis Report

1. Introduction

This report presents an analysis of a sales dataset using **python, pandas, and matplotlib**. The objective is to explore the dataset structure and extract insights regarding sales performance across **regions, products and monthly trends**.

The analysis includes data exploration, aggregation of key metrics, and visualization of results to understand sales distribution and performance patterns.

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2. Dataset Overview

The dataset was loaded from a CSV file and analyzed using pandas, initial exploration included. The primary variables analyzed in this dataset are:

- **Date** – transaction date
- **Region** – sales region
- **Product** – product category
- **Quantity** – units sold
- **Total_Sales** – revenue generated from each transaction

The dataset comprises sales information from following regions:

- **North**
- **South**
- **East**
- **West**

The products mentioned in the dataset are:

- **Phone**
- **Headphone**
- **Laptop**
- **Tablet**
- **Monitor**

This initial exploration ensures the dataset structure is understood before performing analysis.

IMPORTANT INSIGHTS:

Sales amount insights:

- Total revenue from sales: **₹1,23,65,048**
- Average revenue per region sales: **₹1,23,650.48**

Sales quantity insights:

- Total quantity sold: **478 units**
- Average revenue per region sales: **4.78 units**

Product sales revenue insights:

- Headphones: **₹92,268.86**
- Laptop: **₹1,62,050.41**
- Monitor: **₹89,871.40**
- Phone: **₹1,42,969.70**
- Tablet: **₹1,10,936.15**

Product quantity sold insights:

- Headphones: **48 units**
- Laptop: **136 units**
- Monitor: **66 units**
- Phone: **101 units**
- Tablet: **127 units**

3. Region-wise Quantity Analysis

The quantity of products sold across regions was analyzed.

Metrics calculated:

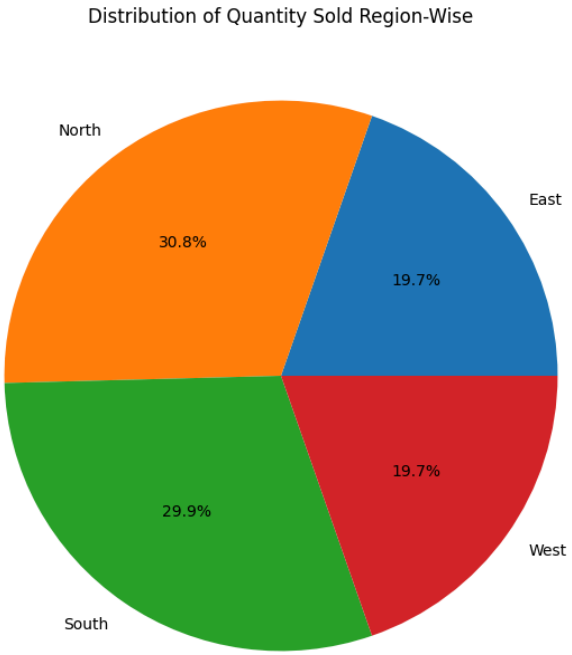
- **Overall quantity sold per region**
- **Average quantity sold per region**

Units sold by region

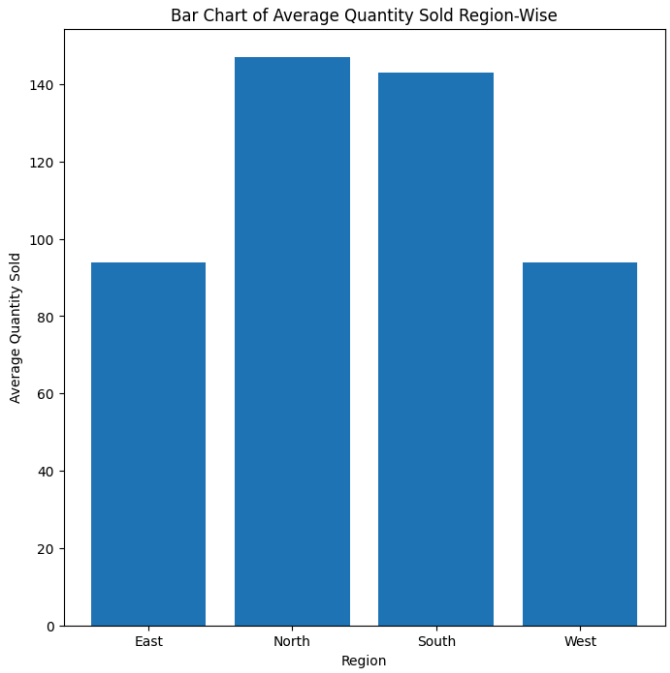
- North: **147 units**
- South: **143 units**
- East: **94 units**
- West: **94 units**

Visualizations created:

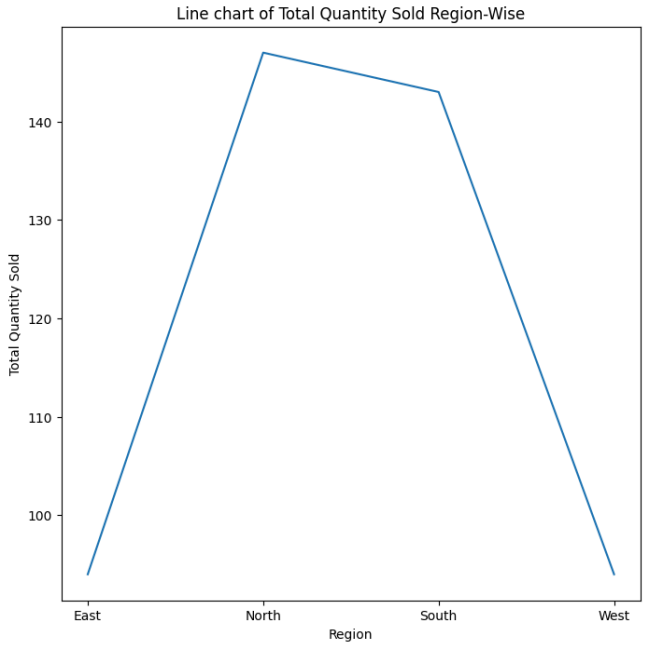
- Pie chart showing **distribution of total quantity sold by region**



- Bar chart comparing **quantity sold across regions**



- Line chart showing **quantity sold across region**



These visualizations highlight which regions contribute the highest number of products sold.

4. Region-wise Sales Analysis

Revenue generated across regions was analyzed.

Metrics calculated:

- **Total sales per region**
- **Average sales per region**

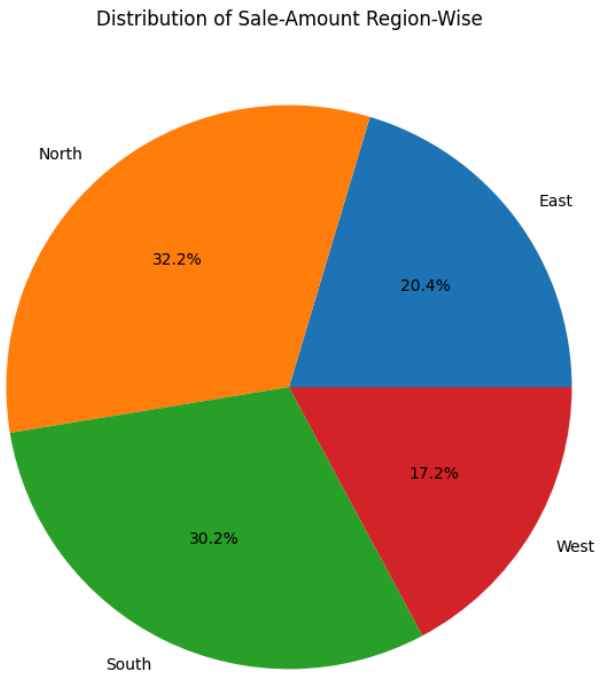
Total sales amount overall : **₹1,23,65,048**

Average sales amount overall: **₹1,23,650.48**

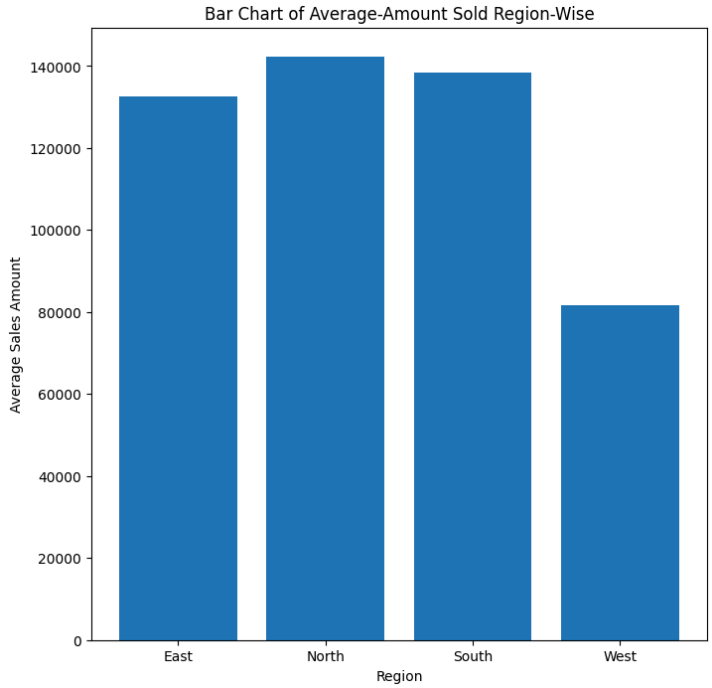
- North: **₹25,19,639 units**
- South: **₹39,83,635 units**
- East: **₹37,37,852 units**
- West: **₹21,23,922 units**

Visualizations created:

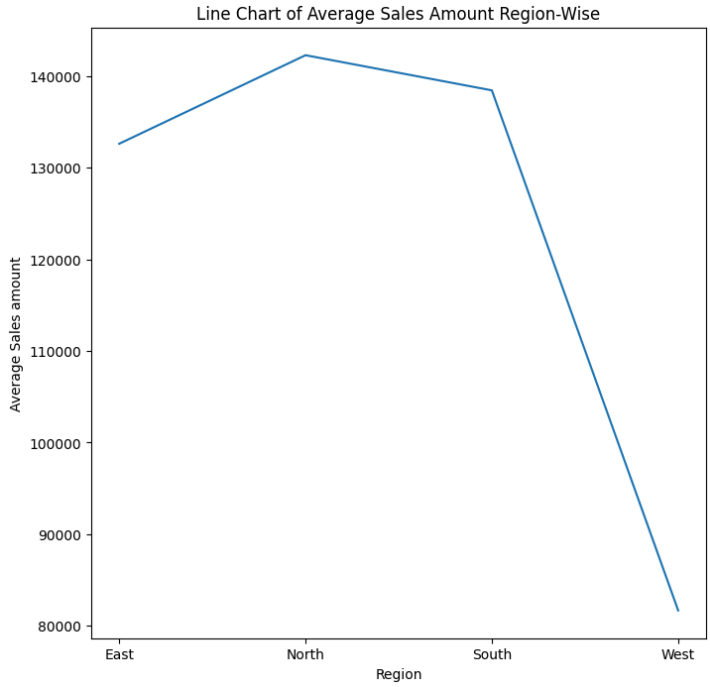
- Pie chart showing **distribution of overall revenue by region**



- Bar chart comparing **average sales by region**



- Line chart illustrating **sales trends by region**



This analysis identifies the regions contributing the largest share of total revenue.

5. Product-wise Quantity Analysis

The quantity of products that were sold was analyzed.

Metrics calculated:

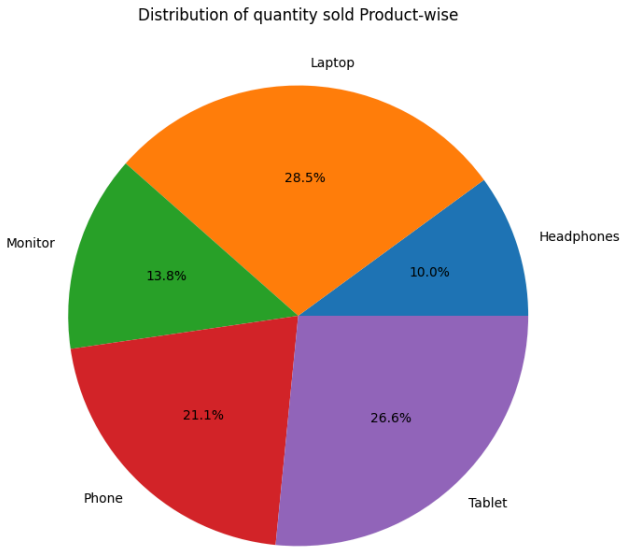
- **Overall quantity sold per product**
- **Average quantity sold per product**

Units sold per product:

- Headphones: **48 units**
- Laptop: **136 units**
- Monitor: **66 units**
- Phone: **101 units**
- Tablet: **127 units**

Visualizations created:

- Pie chart showing **distribution of product quantity sold**

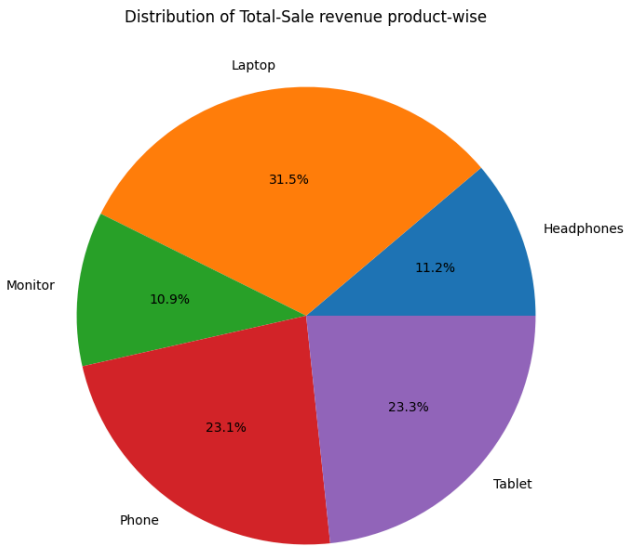


Revenue generated per product:

- Headphones: **₹92,268.86**
- Laptop: **₹1,62,050.41**
- Monitor: **₹89,871.40**
- Phone: **₹1,42,969.70**
- Tablet: **₹1,10,936.15**

Visualizations created:

- Pie chart showing **distribution of product-wise revenue**



Laptop turned out to be the most revenue-generating product

6. Monthly Sales Analysis

This analysis helps identify time based patterns and variations in sales performance across different months.

The **Date** column was converted to a datetime format to enable time-based analysis.

Sales data was then grouped by **month and year** to calculate:

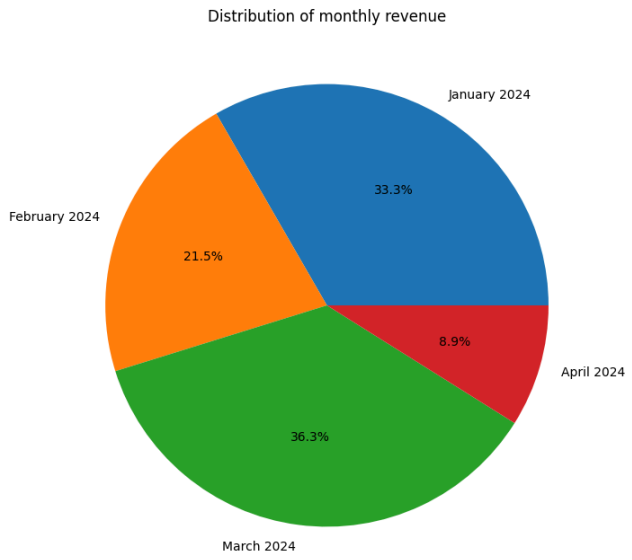
- **Total monthly sales**
- **Product based monthly trends**
- **Overall combined trends**

Total monthly sales:

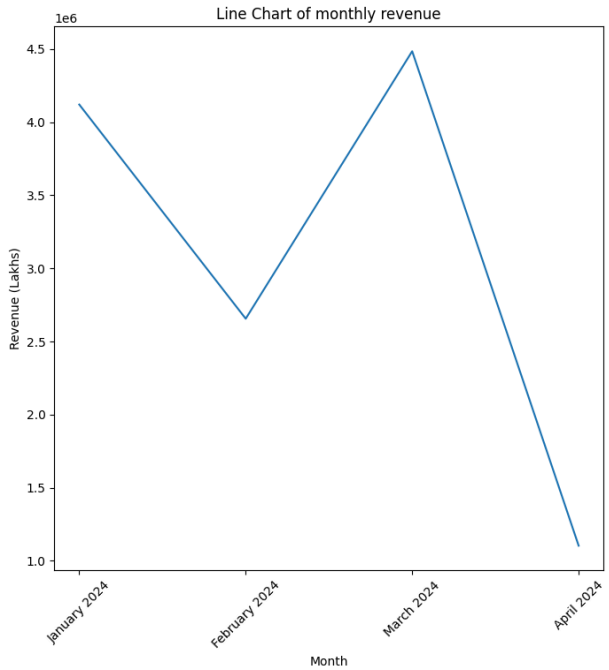
- January 2024 **₹41,20,524**
- February 2024 **₹26,56,050**
- March 2024 **₹44,85,006**
- April 2024 **₹11,03,468**

Visualizations created:

- Pie chart showing **distribution of total sales by month**

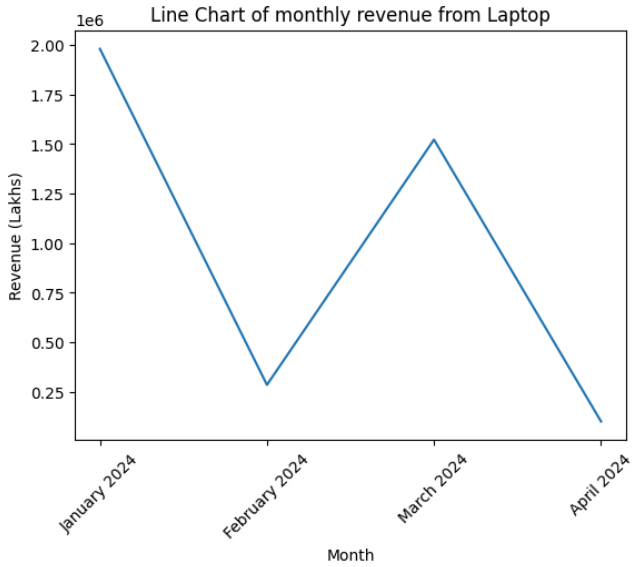


- Line chart illustrating **monthly sales trends over time**

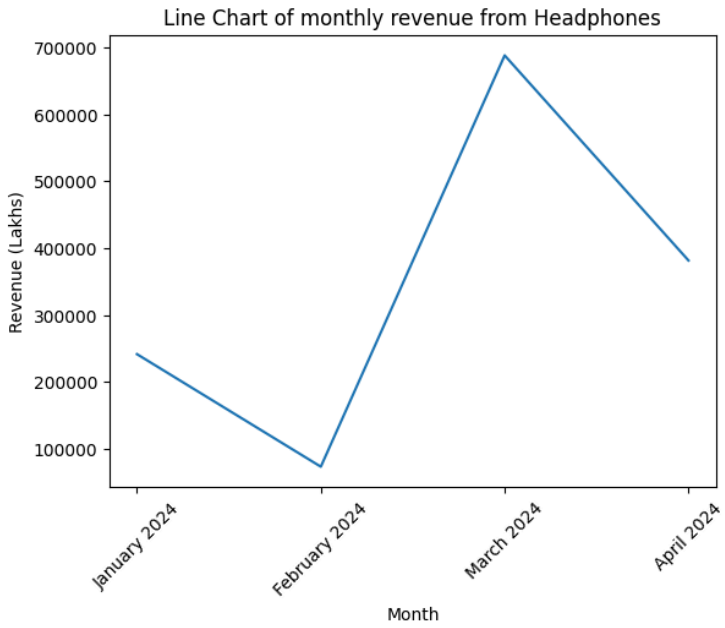


Monthly trends in product revenue:

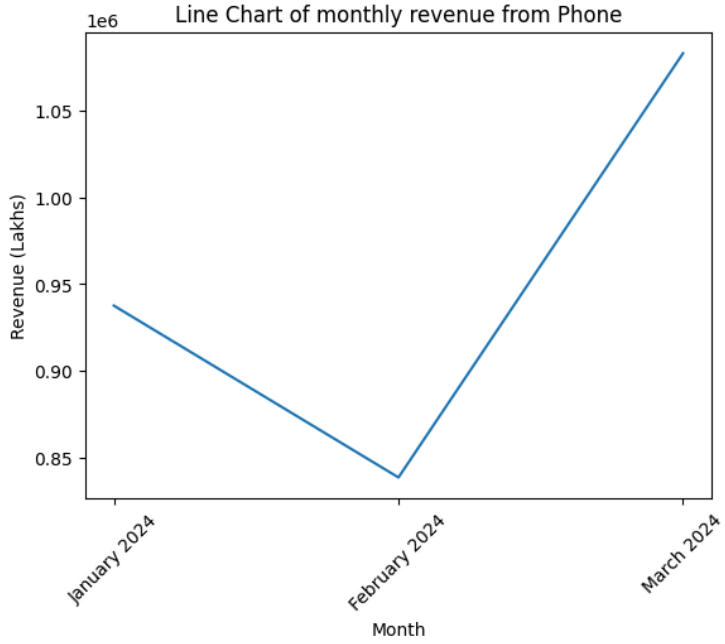
- **Trend in Laptop sales:**



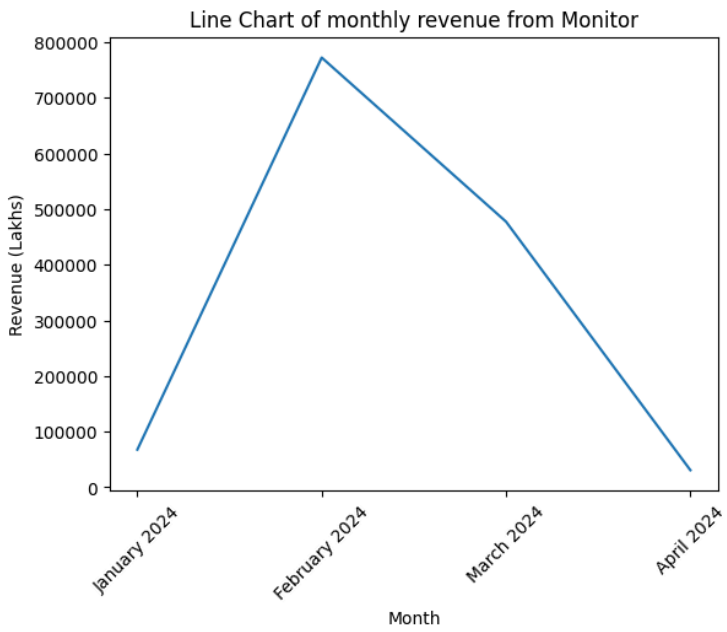
- **Trend in Headphone sales:**



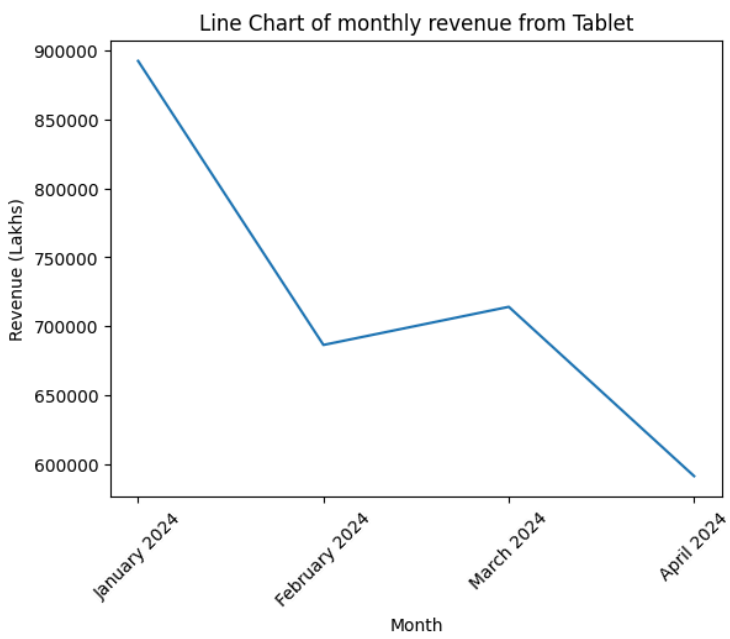
- **Trend in Phone sales:**



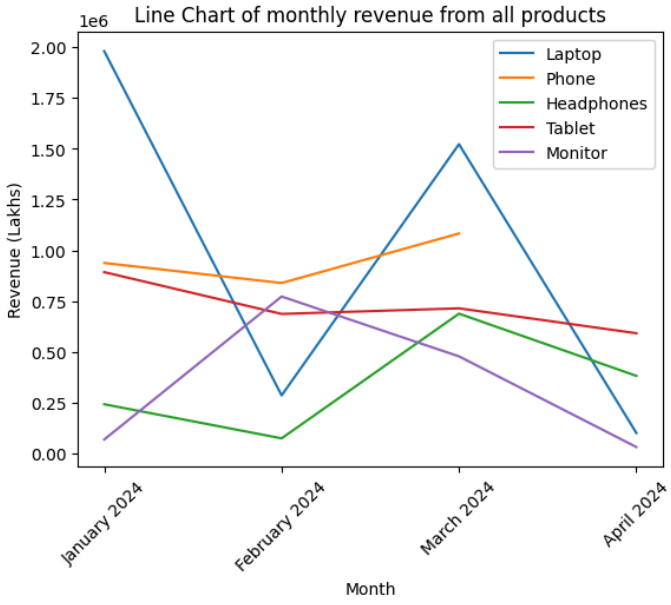
- **Trend in Monitor sales:**



- **Trend in Tablet sales:**



- **Combined trends:**



Key Insights

Important findings from the analysis include:

- Total sales revenue: **₹1,23,65,048**
- Total quantity sold: **478 units**

Regional Performance

- **South** region generated the **highest revenue** of **₹39,83,635**
- **North** region sold the **highest quantity of products of 147 units**

Product Performance

- **Laptop generated the highest revenue** among all products
- **Laptop** also had the **largest sales volume of 136 units**

Monthly Trends

- **March 2024 recorded the highest monthly sales of ₹44,85,006**
- **April 2024 recorded the lowest sales** indicating a drop in performance

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Conclusion

The sales analysis provides insights into how sales vary across **regions, products and time**.

Key findings include:

- Sales performance differs significantly across regions.
- Certain products contribute more to total sales and sales volume.
- Monthly sales trends highlight variations in revenue over time.